Composing Numbers: Part + Part = Whole

Number Bonds & Part-Part-Whole • Standard: CCSS.MATH.1.OA.C.6

BOOK STORY OPENER: Spotting Number Partners at a Glance: Subitizing Domino Pips & Friends of Ten

Your brain has a built-in superpower called subitizing (from the Latin 'subitus', meaning 'sudden'): you can recognize 1 to 5 domino pips instantly without counting one by one! In From Blocks to the Stack, the left half of a domino sits on Shelf Y (Part 1), the right half sits on Shelf X (Part 2), and pressing [+] snaps both halves together into the Whole.

Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 3: Number Bonds, Ch. 8: Missing Addend Mysteries & Appendix B: Station F.2

1. CONCRETE MANIPULATIVE SETUP

Manipulative Kit: Math Domino Studio (12-Frame Tray)

Place the [4|3] domino in the 12-frame tray. Count 4 pips on the left (Part 1) and 3 pips on the right (Part 2).

2. STUDENT PREDICTION (ASK BEFORE PRESSING KEYS)

Combine both parts on the tray: 4 + 3 = 7 total pips.

3. CALCULATOR ACTION (SC-6 KEYCAPS & STACK TRACE)

Keystroke Flow: [4] [ENTER] [3] [+]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect (T, Z, Y, X)
[4]	4	Part 1 (4) enters register X
[ENTER]	4	4 copied up to register Y; represents Part 1
[3]	3	Part 2 (3) enters register X
[+]	7	Plus combines Part 1 (Y) + Part 2 (X) = Whole (7) in X!

4. MATHEMATICAL EXPLANATION & RPN ADVANTAGE

The RPN stack has dedicated registers for parts: register Y holds Part 1, register X holds Part 2, and [+] merges them into the Whole.

Core Investigation Problem:

Pick the [5|4] domino tile. Place it on your 12-frame tray to find the whole. Write the keystrokes.

Decomposing the Whole: Finding the Missing Part

Developing Number Sense • Standard: CCSS.MATH.1.OA.B.4

1. FOLLOW-UP EXTENSION SETUP & PREDICTION

Setup: Place 10 counters in the 12-frame tray. Cover or slide 6 counters to represent known Part 1. Prediction: How many counters remain uncovered in the tray? (10 - 6 = 4).

2. EXTENSION KEYSTROKE SEQUENCE & STACK TRACE

[1] [0] [ENTER] [6] [-]

Key Pressed	Screen (X)	Internal 4-Level Stack Effect
[1]	1	Enter Whole (10)
[0]	10	10 in register X
[ENTER]	10	Whole (10) lifted to register Y
[6]	6	Known Part 1 (6) enters register X
[-]	4	Minus subtracts known Part from Whole: Missing Part is 4!

3. TARGET MISCONCEPTIONS & HANDS-ON BLOCK REMEDIATIONS

Misconception 1: Pressing [+] immediately after the first number without pressing [ENTER] to separate Part 1 and Part 2.

Hands-On Block Remediation: Point to the center divider line on the Math Domino: pressing [ENTER] is the physical divider that pushes Part 1 up to Register Y before Part 2 is typed into Register X.

Misconception 2: Adding the whole and the known part when solving for a missing addend (e.g., computing 10 + 6 instead of 10 − 6).

Hands-On Block Remediation: Place 10 counters in the tray and cover 6; show that the Whole (10) goes into Register Y, the known part (6) enters X, and [−] reveals the hidden part.

4. EXTENSION MATHEMATICAL INSIGHT

Finding a missing addend is inverse addition. Placing the Whole in Y and subtracting known Part in X yields the missing part in X.

Part-Part-Whole & Doubles: The Math Domino Studio

Name: _____ Date: _____ Class: _____

STUDENT REFERENCE GUIDE: HOW DOMINOES FIT THE STACK

- Part 1 lives in register Y. Part 2 enters register X.
- Pressing [+] combines both parts into the Whole in register X.
- To find a missing part: start with Whole in Y, enter Part 1 in X, and press [-]!

Key Sequence: `4` `ENTER` `3` `+`

Problem 1: Make 6: Part 1 is 2. What is Part 2? Write the keystrokes.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 2: Part 1 is 7, Part 2 is 2. What is the Whole?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 3: Three parts make a whole: 2, 3, and 4. Add them on the stack.

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Problem 4: The Whole is 10. Part 1 is 7. What is Part 2?

1. Block Model / Sketch:

2. Mathematical Strategy & Work:

3. Prediction: _____

Final Answer: _____

[ ] Checked on SC-6

Unit 02: Part-Part-Whole & Doubles: The Math Domino Studio — Answer Key

Complete solutions and SC-6 verification keystrokes for Grades K–1

1. CORE & EXTENSION INVESTIGATION SOLUTIONS

Core Investigation: Pick the [5|4] domino tile. Place it on your 12-frame tray to find the whole. Write the keystrokes.

Core Solution: 9 pips. Keystrokes: [5] [ENTER] [4] [+]

Core Check Keys: 5 ENTER 4 +

Extension Challenge: The whole is 8. Part 1 is 5. Find the missing Part 2 using the calculator.

Extension Solution: Missing Part is 3. Keystrokes: [8] [ENTER] [5] [-]

Ext. Check Keys: 8 ENTER 5 -

2. STUDENT LAB WORKSHEET SOLUTIONS (PROBLEMS 1–4)

Problem 1: Make 6: Part 1 is 2. What is Part 2? Write the keystrokes.

Expected Answer: 4

SC-6 Verification Keys: 6 ENTER 2 -

Problem 2: Part 1 is 7, Part 2 is 2. What is the Whole?

Expected Answer: 9

SC-6 Verification Keys: 7 ENTER 2 +

Problem 3: Three parts make a whole: 2, 3, and 4. Add them on the stack.

Expected Answer: 9

SC-6 Verification Keys: 2 ENTER 3 + 4 +

Problem 4: The Whole is 10. Part 1 is 7. What is Part 2?

Expected Answer: 3

SC-6 Verification Keys: 1 0 ENTER 7 -

Hands-On Block & Stack Verification Protocol

Students first build the physical block model and write their mathematical strategy and prediction before verifying their result on the StackCalc calculator.